

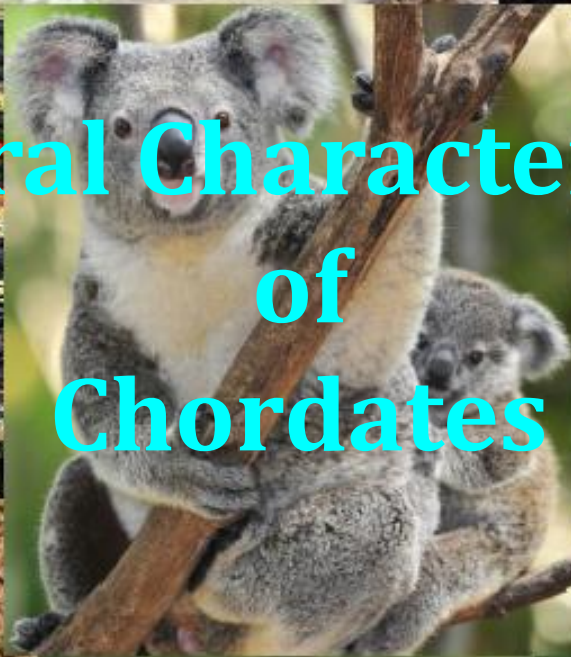


ZOO 1302 Vertebrate Diversity

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Lesson 1



General Characteristics of Chordates

Learning Outcomes

At the end of the lesson you would be able to describe the

- general characteristics of chordates.
- chordate hallmarks.

Phylum – Chordata

- Most complex of all phyla.
- Last major group of the Animal Kingdom – created by Balfour (1880)
- Chordates (Chordata) are a group of animals that includes more than 75,000 living species.
- The name of the phylum derives from - *notochord*
- *Noton* – back + *chordata* – cord

- What is the difference between Invertebrates and Chordates ?
- All members of the phylum possess a notochord.
- How does the notochord persist in chordates?
- Largest of the duterostome phyla.
- highest and the most important phylum comprising a vast variety of ling and extinct animals including man.

Diversity of chordates

- Exhibit an astonishing diversity of form, physiology and habit.

1. Numerical strength

- Number of chordate species is not usually large.



- Of the sub phylum Vertebrates under phylum chordata most numerous species - -----
- *Highest to lower*

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2. Size

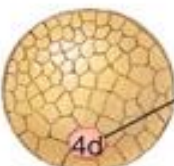
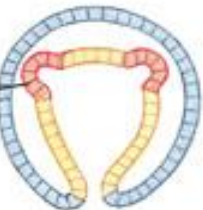
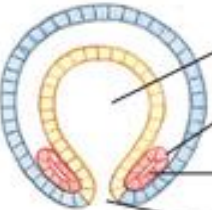
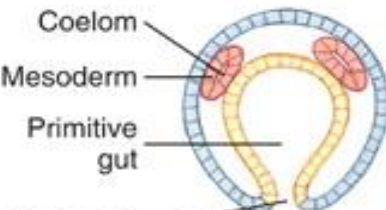
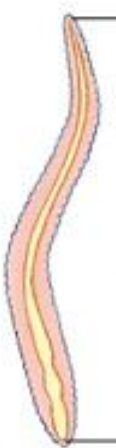
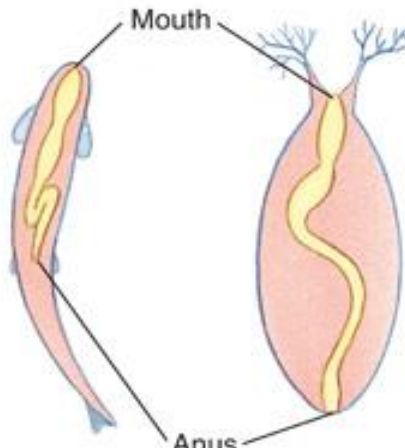
- medium to large in size
- *Largest animal/mammal in the world*
- *Largest fish*
- *Largest animal/mammal – terrestrial*

3. Ecology

- ecologically most successful
- able to occupy most kinds of habitats.
- they have adapted themselves to more modes of existence than other groups including the arthropods.
- found in sea, freshwater, air, from poles to equator.
- all lower chordates –
- fish –
- Birds -
- some amphibians, reptiles and mammals -
- most tunicates (lower chordate) -
- But other chordates -
- parasitic -

Characteristics of Phylum Chordates

- bilaterally symmetrical
- three germ layers - triploblastic
- well-developed coelom
- deuterostomes
 - invertebrates like Echinodermata
eg. sea stars sand dollars, sea urchins and Hemichordata eg. acorn worms also deuterostomes.

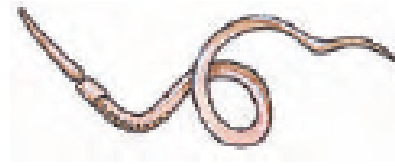
PROTOSTOMES	DEUTEROSTOMES
 Spiral cleavage Cleavage mostly spiral	Cleavage mostly radial  Radial cleavage
 Cell from which mesoderm will derive Endomesoderm usually from a particular blastomere designated 4d	Endomesoderm from enterocoelous pouching (except vertebrates) Endomesoderm from pouches from primitive gut 
 Primitive gut Mesoderm Coelom Blastopore In coelomate protostomes the coelom forms as a split in mesodermal bands (schizocoelous)	All coelomate, coelom from fusion of enterocoelous pouches (except vertebrates, which are schizocoelous)  Coelom Mesoderm Primitive gut Blastopore
 Anus Annelid (earthworm) Mouth Mouth forms from or near blastopore; anus a new formation Embryology mostly determinate (mosaic) Includes phyla Platyhelminthes, Nemertea, Annelida, Mollusca, Arthropoda, Chaetognatha, Phoronida, Ectoprocta, Brachiopoda, minor phyla	Anus forms from or near blastopore, mouth a new formation Embryology usually indeterminate (regulative) Includes phyla Echinodermata, Hemichordata, Chordata  Mouth Anus

Deuterostomia

Hemichordata



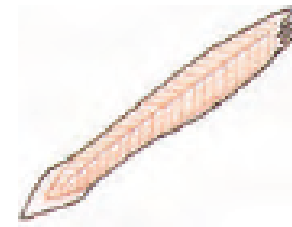
Echinodermata



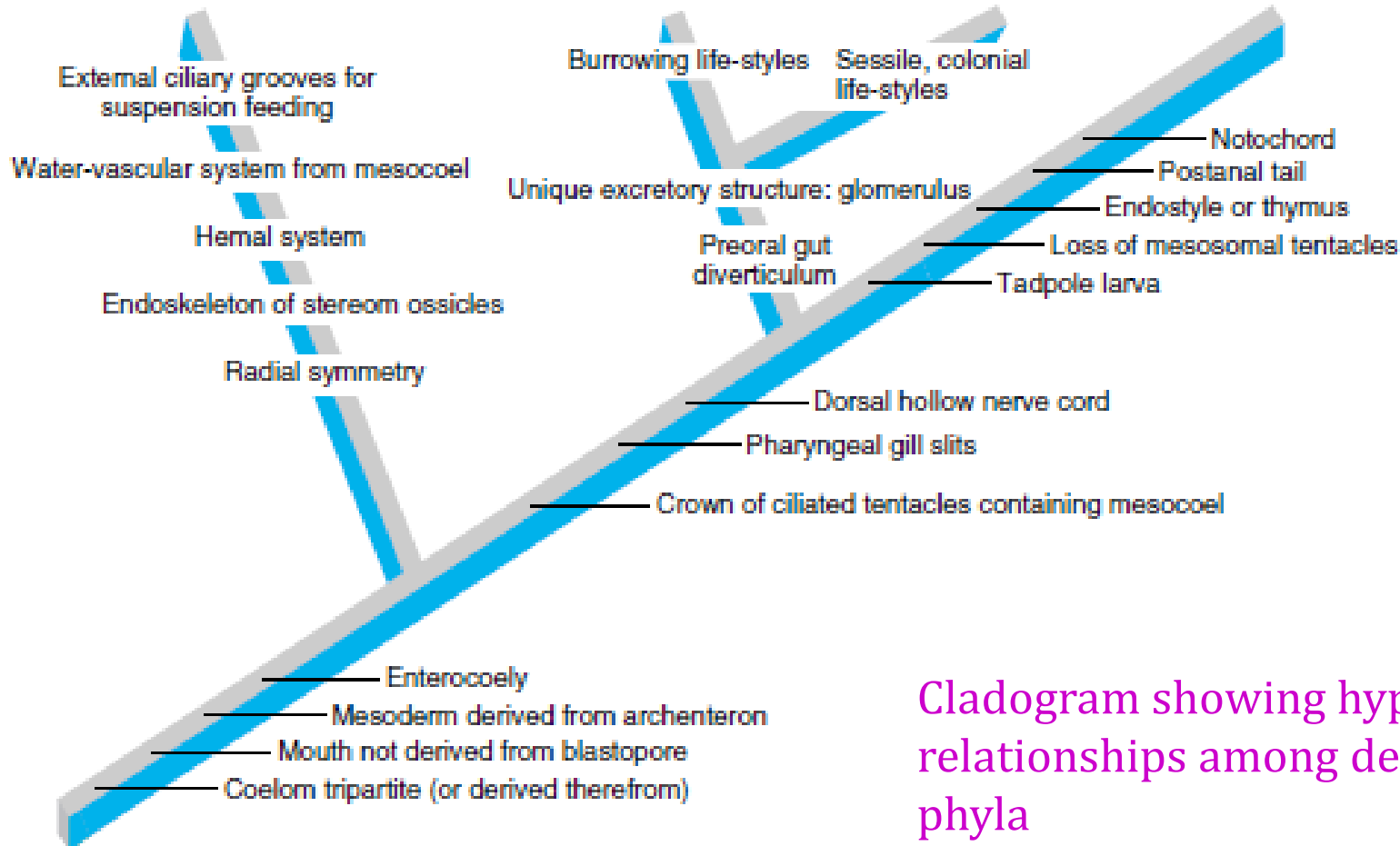
Enteropneusta



Pterobranchia



Chordata



Cladogram showing hypothetical relationships among deuterostome phyla

- presence of notochord
- single, dorsal tubular nerve cord
- pharyngeal pouches - present in some stage in the life cycle.
- endostyle in floor of pharynx or a thyroid gland derived from the endostyle
- postanal tail
- segmented muscles

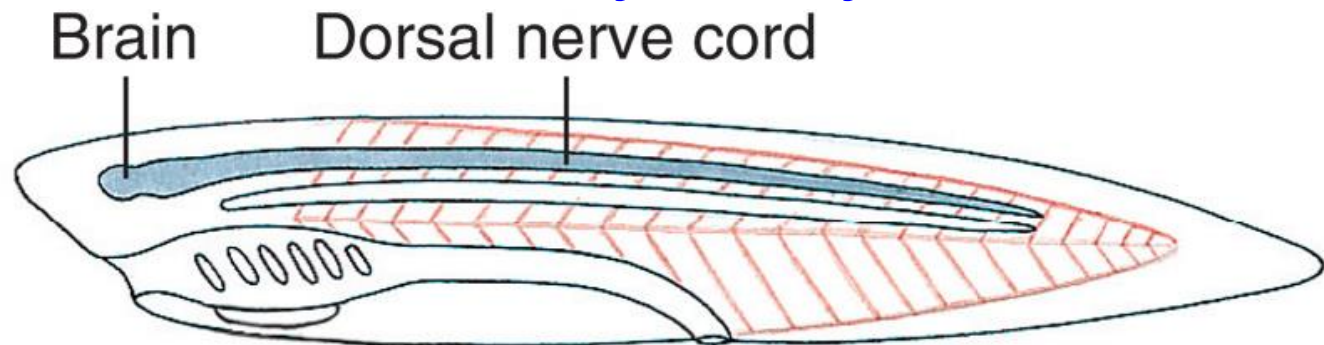
- ventral heart with ventral and dorsal blood vessels
- complete digestive system
- cartilaginous or bony endoskeleton present in the majority of members
- ***segmentation***– if present restricted to outer body wall, head, and tail and ***not extending*** into coelom

Five Chordate Hallmarks

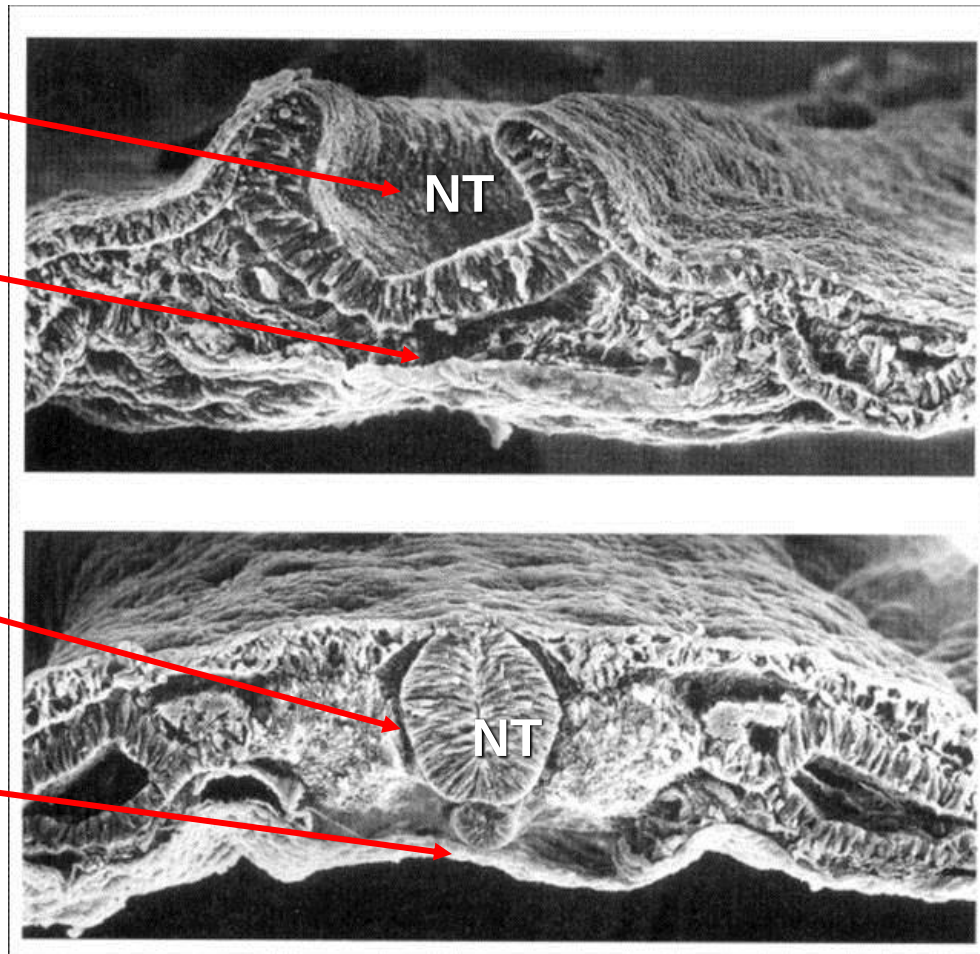
- There are *five* characters that set chordates apart from all other phyla.
- *Five chordate hallmarks.....*
 1. Dorsal tubular nerve cord
 2. Notochord
 3. Pharyngeal pouches and slits
 4. Endostyle or Thyroid gland
 5. Postanal tail
- These characteristics are always found at some embryonic stage.

1. Dorsal tubular nerve cord

- dorsal to the notochord and alimentary canal (invertebrates – ventral)
- single tubular structure (invertebrates – solid)
- anterior end of nerve cord – enlarged to form brain in vertebrates
- protected by the vertebral column of vertebrates
- the anterior brain is surrounded by a bony or cartilaginous case

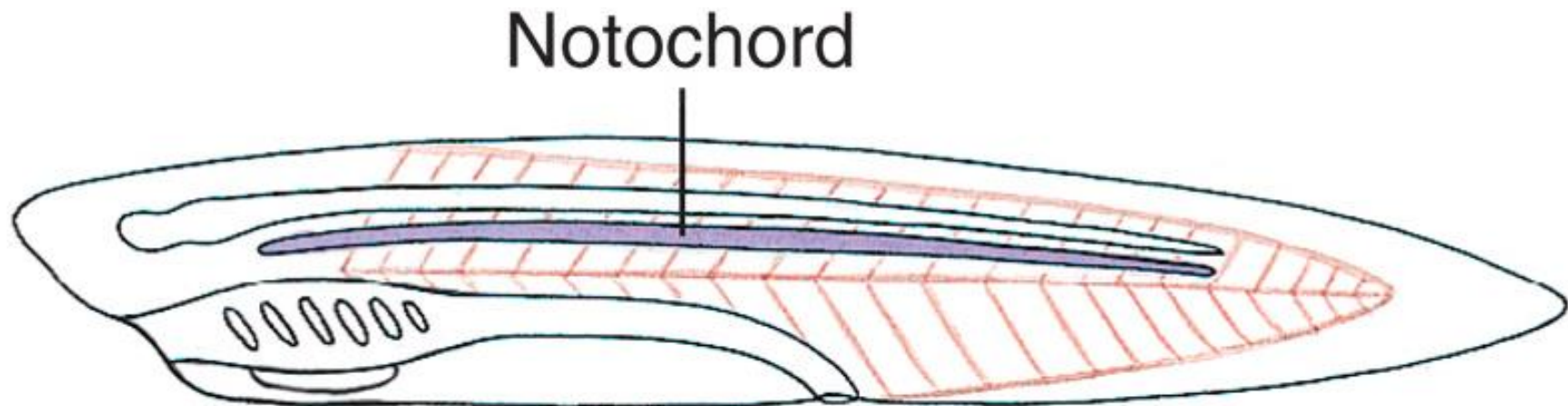


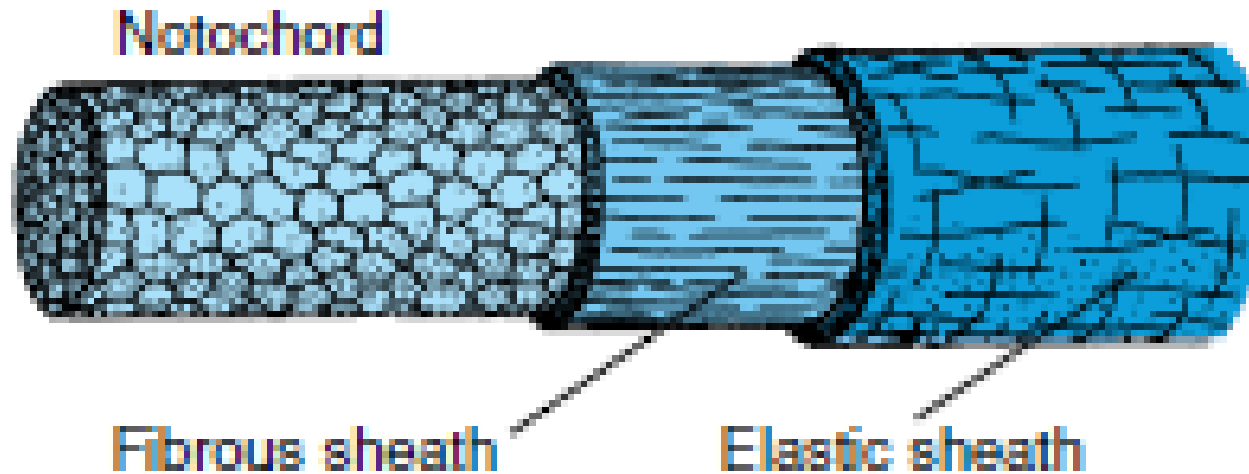
Notochord & Formation of Neural Tube



2. Notochord

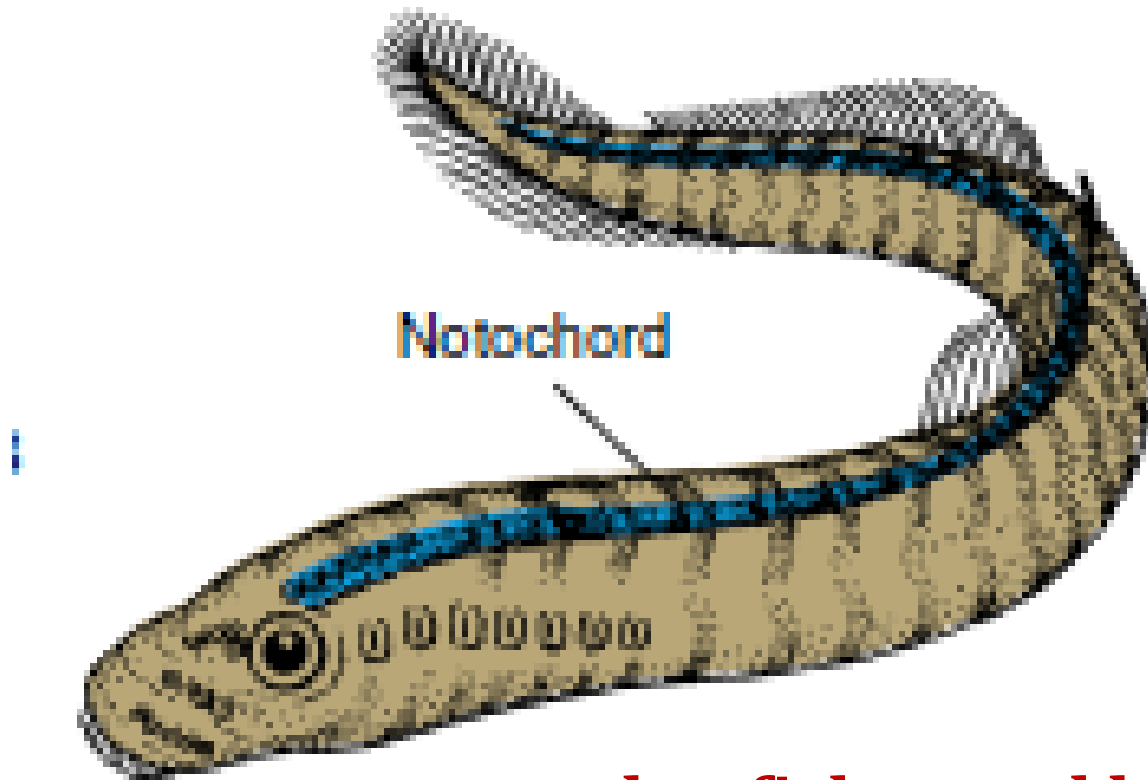
- flexible, rod-like structure – derived from mesoderm
- extends length of body - ventral to nervous system's spine
- **first part** of the endoskeleton to appear in embryo
- bend without shortening – permits undulatory movements of the body





- Notochord is a primitive type of endoskeleton which is characteristic of all chordates at some stage of the life cycle.
- Cells of the notochord are thick walled, pressed together closely, and filled with semifluid.
- Stiffness is caused mainly by turgidity of fluid-filled cells and surrounding connective tissue sheaths.

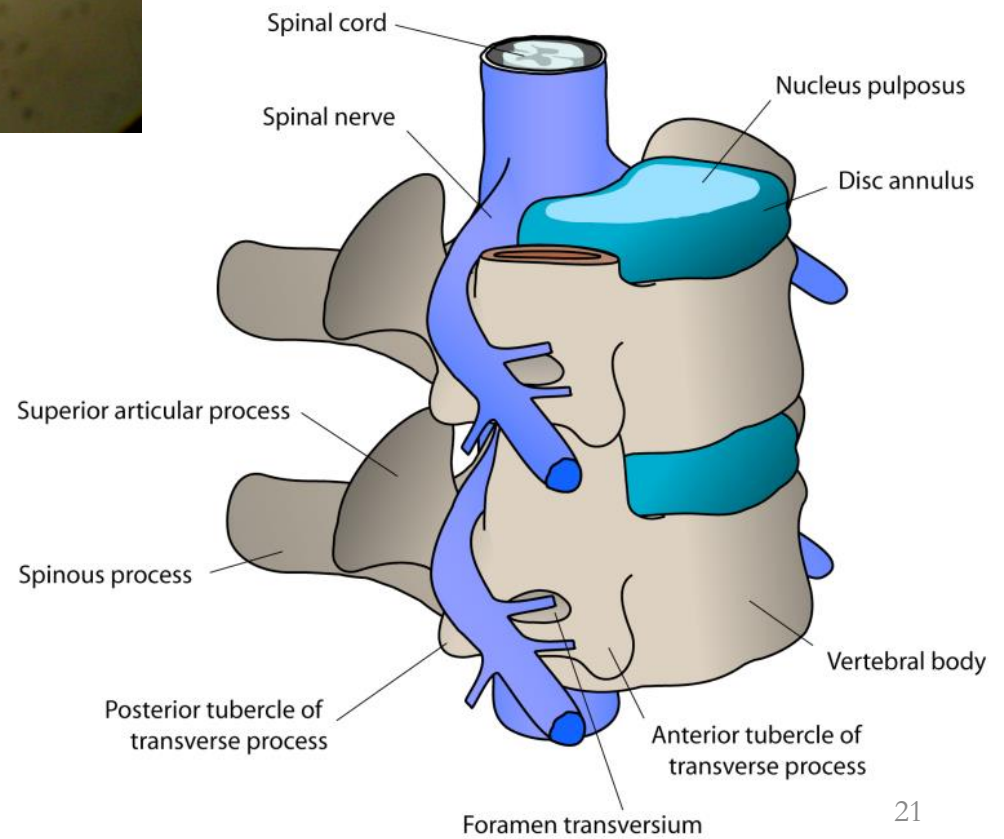
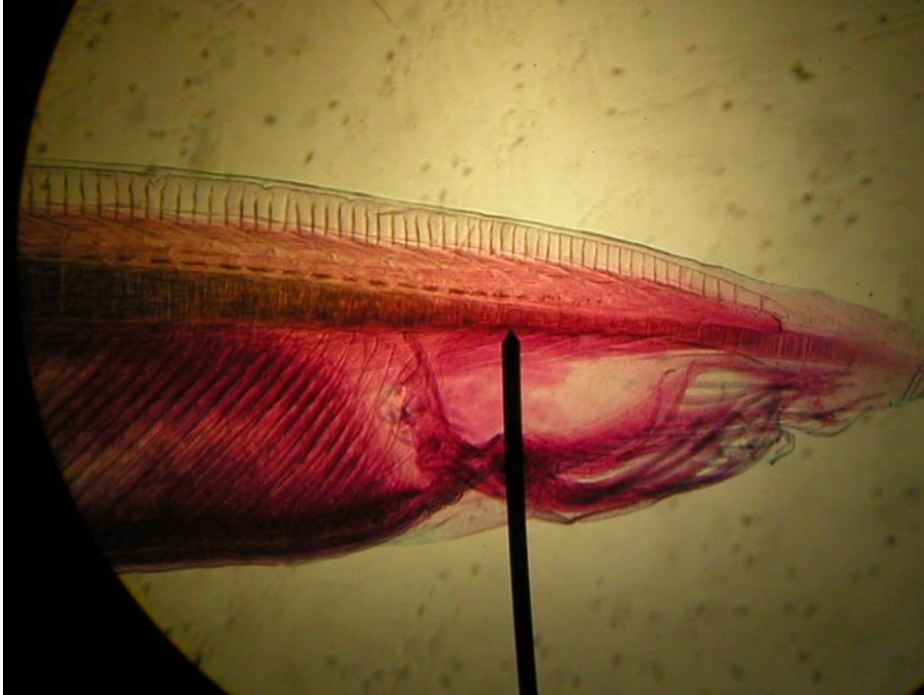
- In most protochordates and in jawless vertebrates the notochord persists throughout life.
- But in other vertebrates it is largely replaced by vertebrae.



hagfishes and lampreys

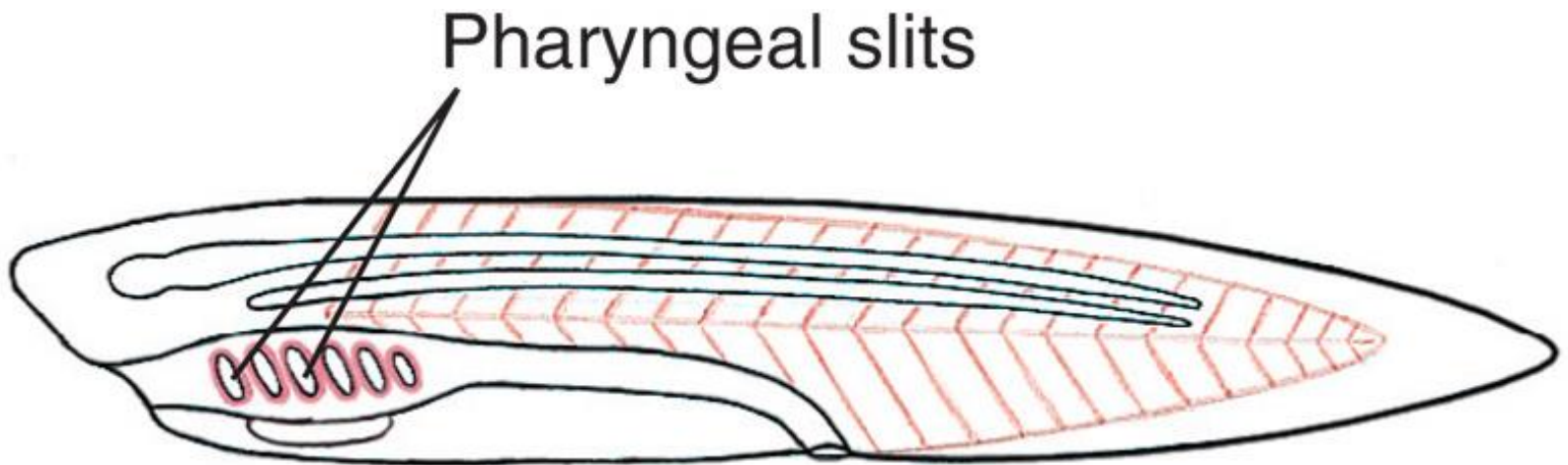
■ *The notochord provides ...*





3. Pharyngeal pouches and slits

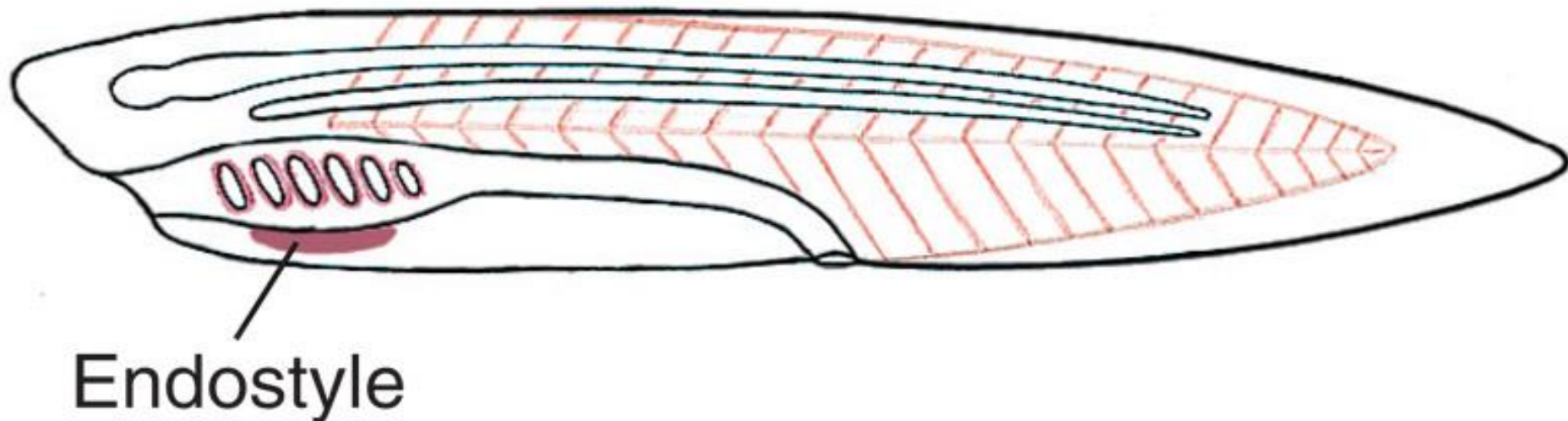
- Pairs of opening through the pharynx
- Pharyngeal slits - openings that lead from the pharyngeal cavity to outside
- Perforated pharynx – as a filter feeding apparatus in protochordates



- Lead to evolution of internal gills – a respiratory organ in aquatic vertebrates
- In reptiles, birds, and mammals the gill slits are vestiges, occurring only in the embryo

4. Endostyle or Thyroid gland

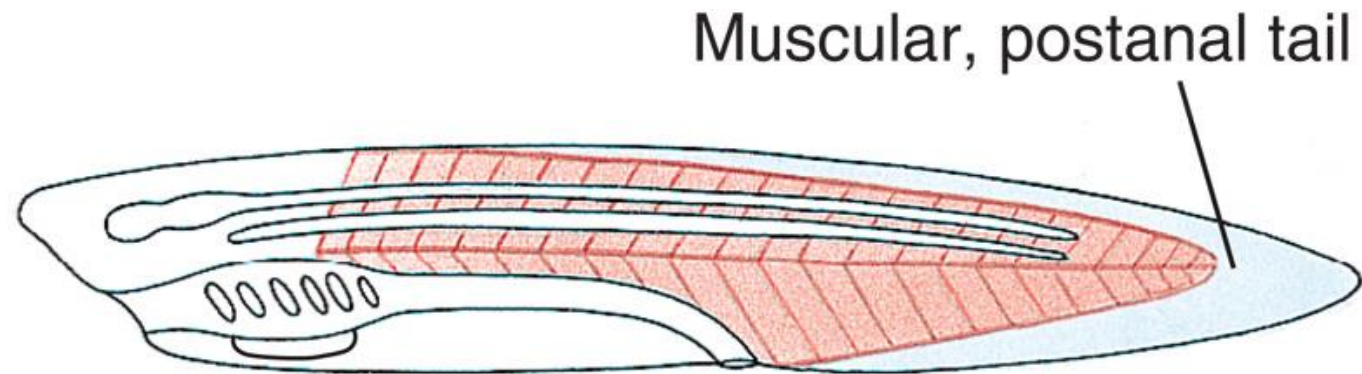
- The endostyle in the pharyngeal floor, secretes mucus that traps food particles
- Thyroid gland is a derivative of endostyle - found in all chordates – not present in other animals



- Found in protochordates and lamprey larvae - some cells - secretes iodinated proteins.
- Homologous to the iodinated-hormone-secreting thyroid gland in adult lampreys and other vertebrates.

5. Postanal tail

- The postanal tail, along with somatic musculature and the stiffening notochord, provides motility in larval tunicates and amphioxus.
- As it is added behind the end of the digestive tract - evolved for propulsion in water.
- Evident in humans – as a vestige – the coccyx
- Other mammals – waggable tail





This is a sagittal view anatomical illustration of the human spine. The vertebrae are shown in a yellowish-tan color, with blue intervertebral discs. The sacrum is a large, triangular bone at the base of the spine, and the coccyx is a small, tail-like structure at the very bottom. Two black lines with labels point to these structures.

Sacrum

Coccyx

Phylum - Chordata divided into three basic groups:

1. Sub phylum - Urochordata (tunicates)
 2. Sub phylum - Cephalochordata (lancelets)
 3. Sub phylum – Vertebrata (Craniates)
- Of these, the **vertebrates** - lampreys, mammals, birds, amphibians, reptiles and fishes—are the most familiar and are the group to which humans belong.

Protochordates

Ancestry and Evolution

- Zoologists at first speculated that chordates evolved within the *protostome lineage - annelids and arthropods*.
- But they realized that supposed morphological similarities were not homologous – *discarded the idea*.
- Early in the 20th century - *chordates must have originated within the deuterostome branch (which includes echinoderms, hemichordates and chordates) of the animal kingdom*.
- **Deuterostomians** – *common origin in ancient Precambrian seas*.

- At the base of the Cambrian period 570 million years ago - *the first distinctive chordates arose from a lineage related to echinoderms and hemichordates.*
- Evidence from phylogenetic analysis of *gene sequences, development, and morphology* strongly suggest that a clade containing both echinoderms and hemichordates (half chordates) is the sister group of chordates.
- Information about the biology of the earliest chordates can be gathered from examination of the two living chordate groups that are not vertebrates - *Urochordata and Cephalochordata.*

Jurassic	200-145 mya First bird fossils.
Triassic	250-200 mya First mammalian fossils.
Permian	300-250 mya
Carboniferous	360-300 mya First reptile fossils.
Devonian	415-360 mya First tetrapod fossils (amphibians).
Silurian	445-415 mya First jawed fish fossils.
Ordovician	490-445 mya
Cambrian	540-490 mya First fossils of species belonging to the subphylum cephalochordata and jawless fish of the subphylum vertebrata.

mya = million years ago

- Chordate evolution, like that of arthropods - *includes major habitat transitions.*
- The earliest chordates were all marine animals - *tunicates and lancelets.*
- As chordates continued to evolve - *they spread to freshwater habitats and ultimately to land.*
- The amphibians - *represent an intermediate phase in the water to land transition of chordates.*
- The evolution of birds - *further increased the distribution of chordates by expanding their populations into aerial habitats.*

- The evolution of chordates led to a variety of specialized *structural features* and *behavioral modifications*.
- Appeared when environmental pressures imposed on different groups within the phylum.
- Vertebrates show an enormous range of features that represent different evolutionary modifications of the same original structure.
Eg. bird beak

THE END

